习题5.

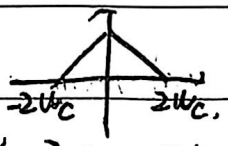
5-1. (1) $\omega=\frac{2\pi}{T}=8\times10^3\pi>2\times10^3\pi\times2$，可以

(2). $\omega=\frac{2\pi}{T}=2\times10^3\pi<2\times10^3\pi\times2$，不可以

(3) $\omega=\frac{2\pi}{T}=4\times10^3\pi=2\times10^3\pi\times2$，可以

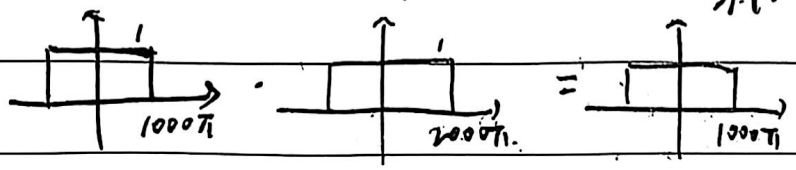
(4). $\omega=\frac{2\pi}{T}=4\times10^4\pi>2\times10^3\pi\times2$，可以

5-2. (1) $2\omega_m=2\times3000\pi=6\times10^3\pi$

(3). $x(t)=\left(\frac{\sin\omega_c t}{\pi t}\right)^2 \xrightarrow{F}$ 

$2\omega_m=2\times2\omega_c=4\omega_c$

(4) $x(t)=\left(\frac{\sin1000\pi t}{\pi t}\right)*\left(\frac{\sin2000\pi t}{\pi t}\right)$



$2\omega_m=2000\pi$.

5-4

$$x_r(t)=\sum_{n=-\infty}^{+\infty}x(nT)h(t-nT)$$

$$=\sum_{n=-\infty}^{+\infty}x(nT)\cdot\delta(t-nT)*h(t)$$

$$X_r(j\omega)=X_p(j\omega)\cdot H(j\omega)$$

$\because H(j\omega)=0.2\,Sa^2(0.1\omega)$

$\because x(t)=\cos2\pi t$, $X(j\omega)=\pi[\delta(\omega+2\pi)+\delta(\omega-2\pi)]$

$$\cancel{X_p(j\omega)=5\sum_{k=-\infty}^{+\infty}\pi[\delta(\omega-k10\pi+2\pi)+\delta(\omega-10k\pi-2\pi)]}$$

则 $X_r(j\omega)=\pi\sum_{k=-\infty}^{+\infty}[\delta(\omega-10k\pi+2\pi)+\delta(\omega-10k\pi-2\pi)]\cdot Sa^2(0.1\omega)$

则 $x_r(t)=\sum_{k=1}^{\infty}\left[Sa^2(k\pi+0.2\pi)\cos(10k+2)\pi t+Sa^2(k\pi-0.2\pi)\cos(10k-2)\pi t\right]$
$+Sa^2(0.2\pi)\cos2\pi t$.

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5-6.

(1). $f(t)=x_1(t)+x_2(t)$.

$\omega_M=300Hz$. 则 $T_{min}=\frac{1}{2\times300Hz}=\frac{1}{600}S$

(2). $f(t)=x_1(t/2)$.

$\omega_M=50Hz$. 则 $T_{min}=\frac{1}{2\times50Hz}=0.01S$

(3). $f(t)=x_2(2t)$

$\omega_M=600Hz$. 则 $T_{min}=\frac{1}{2\times600Hz}=\frac{1}{1200}S$.

5-7 $X(j\omega)=2\pi A\delta(\omega)+\frac{B\pi}{2}[e^{j\theta}\delta(\omega-\frac{2\pi}{T})+e^{-j\theta}\delta(\omega+\frac{2\pi}{T})]$

$x_p(t)=x(t)p(t)=x(t)\sum_{n=-\infty}^{+\infty}\delta[t-n(T+\Delta)]$

$X_p(j\omega)=\frac{1}{T+\Delta}X(j(\omega-k\omega_s))$, $\omega_s=\frac{2\pi}{T+\Delta}$

因为 $H(j\omega)=\begin{cases}1 & |\omega|<\frac{\pi}{T+\Delta}\\ 0, & 其他\end{cases}$ 故 $\frac{\pi}{T+\Delta}>\frac{2\pi\Delta}{T(T+\Delta)}$

$\Delta>\frac{T}{2}$

$Y(j\omega)=2\pi A\delta(\omega)+\frac{Be^{j\theta}\pi}{2}\delta(\omega-\frac{2\pi\Delta}{T(T+\Delta)})+\frac{Be^{-j\theta}\pi}{2}\delta(\omega+\frac{2\pi\Delta}{T(T+\Delta)})$

则 $y(t)=A+B\cos[\frac{2\pi\Delta t}{T(T+\Delta)}+\theta]$

$x(at)=A+B\cos(\frac{2a\pi}{T}t+\theta)$

故 $\frac{2\pi\Delta}{(T+\Delta)\cdot T}=\frac{2a\pi}{T}$.

得 $a=\frac{\Delta}{T+\Delta}$

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5-9 $\omega_m = \frac{3\pi}{7}$，则 $2\omega_m = \frac{6}{7}\pi$.

$N < \frac{2\pi}{\frac{6}{7}\pi} = \frac{7}{3}$.

故 $N = 2$.

5-12 $x_d[n] = x_c(nT)$

$T = 10^{-3}s$.

$\omega_m = 1000\pi$，则 $\omega_s = 2000\pi$.

(1) $X_c(j\omega)$ 也为实函数.

(2). $X_c(j\omega)_{max} = T = 10^{-3}$.

(3). $X_c(j\omega) = 0$，$250\pi \le |\omega| \le 1000\pi$

(4) $X_c(j\omega) = X_c(j(\omega - 1000\pi))$.

5-18. $y[n] = \frac{1}{3}y[n-1] + x[n]$. $Y(j\omega) = \frac{1}{3}e^{-j\omega}Y(j\omega) + X(j\omega)$

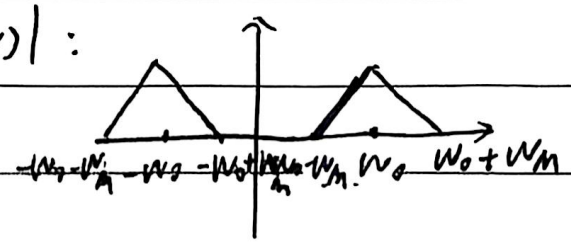
$H_d(e^{j\omega}) = \frac{1}{1-\frac{1}{3}e^{-j\omega}}$.

则 $H_c(j\omega) = \frac{1}{1-\frac{1}{3}e^{-j\omega T}}$.

5-22. (1). $z(t) = \cos\omega_0 t \cdot x(t)$

$= \frac{1}{2}x(t)\cdot e^{-j\omega_0 t} + \frac{1}{2}x(t)e^{j\omega_0 t}$.

则 $Re|Z(j\omega)|$：



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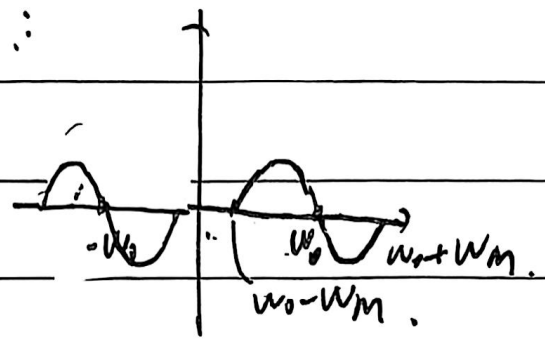
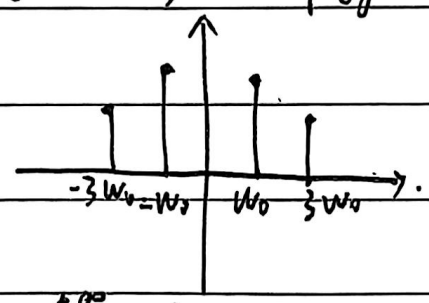
$Im|z(jw)|$:

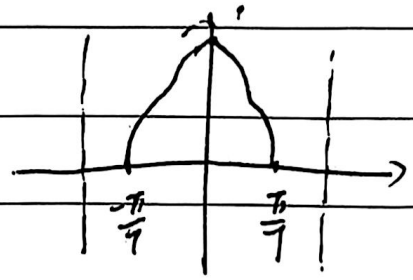


$p(t)=sgn(\cos w_0 t)$. $\quad P(jw)=\sum_{k=1}^{\infty}\frac{-4j}{\pi(2k-1)}[\delta(w-(2k-1)w_0)-\delta(w+(2k-1)w_0)]$



$Y(jw)=\frac{1}{\pi}\sum_{k=1}^{+\infty}\frac{1}{2k-1}[X(j(w-2kw_0))+X(j(w+2kw_0))]$

(2). $H(jw)=\begin{cases}0, & |w|>w_m\\ \frac{\pi}{2}, & |w|\le w_m\end{cases}$



5-24. (1). $X_p(jw)=\frac{1}{T}\sum_{k=-\infty}^{+\infty}X(j(w-kw_s))$.

$y(t)=\int(x_p(t)-x_p(t-\Delta))dt$. $\quad Y(jw)=\frac{X_p(jw)}{jw}(1-e^{-jw\Delta})$.

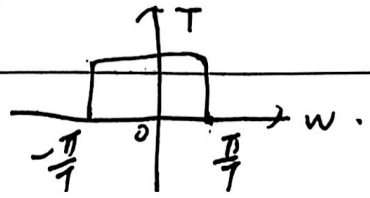
(2). $1-e^{-jw\Delta}\ge 0$. 且 $\Delta\le T$. $\qquad |w|\le\frac{\pi}{T}$

故 $\Delta_{max}=T$.

(3). $X(jw)=H_m(jw)\cdot Y(jw)$. $\qquad$ 则 $H_m(jw)=\begin{cases}T\frac{jw}{1-e^{-jw\Delta}}, & \\ 0, & |w|>\frac{\pi}{T}\end{cases}$

$H_m(jw)=\frac{jw}{1-e^{-jw\Delta}}$, $|w|\le\frac{\pi}{T}$

故理想低通滤波器:



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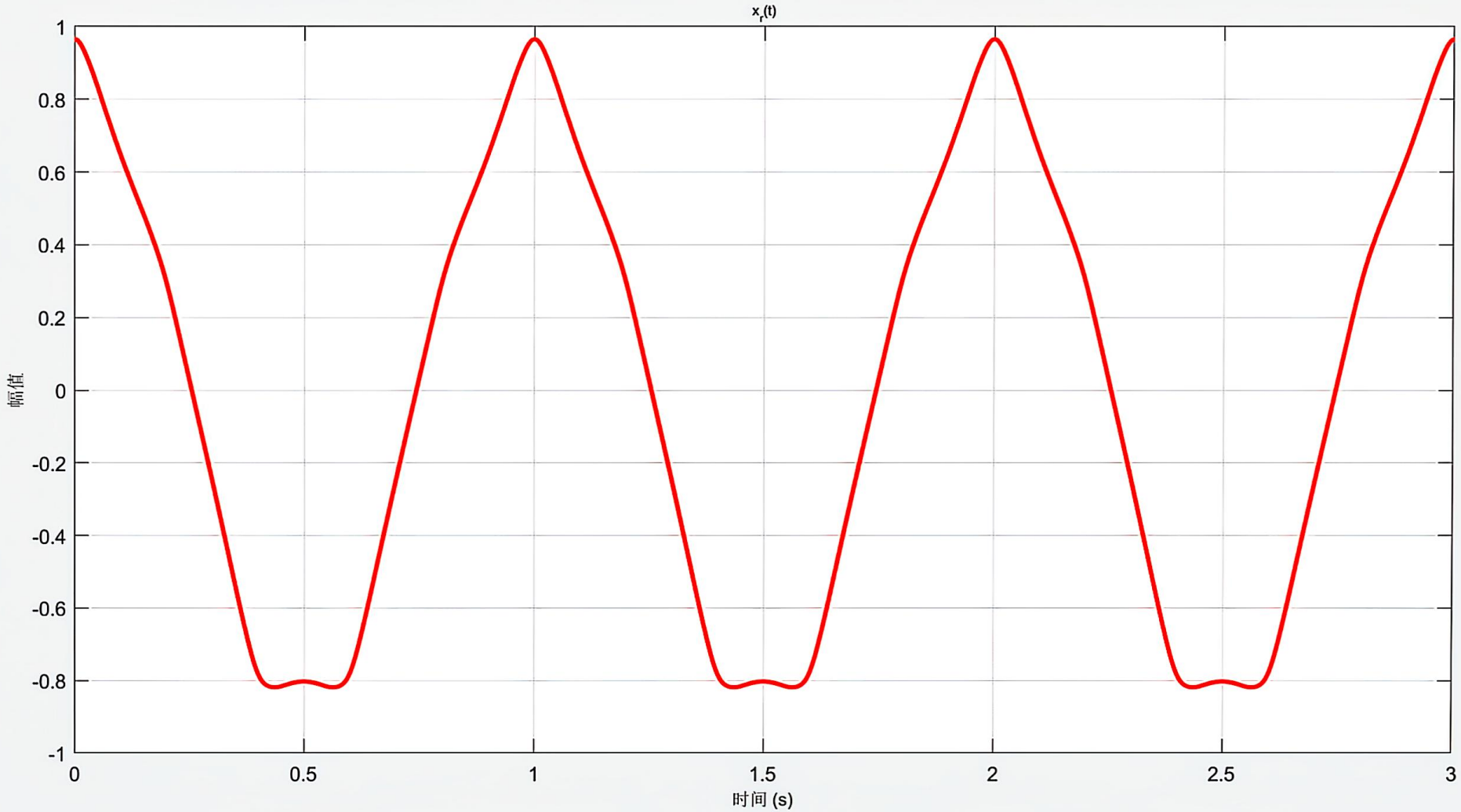

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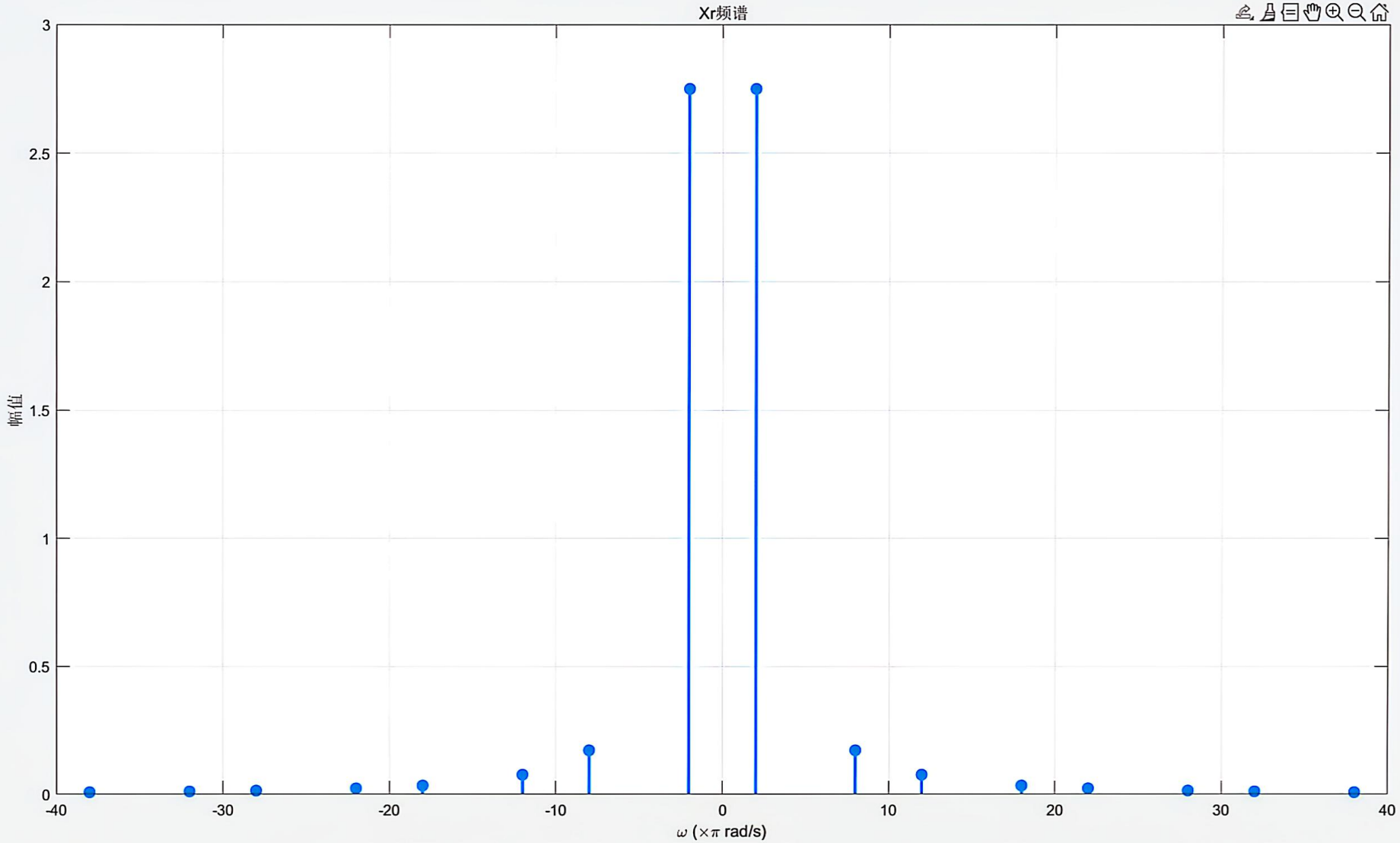

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